

Algebra

$\sin n\pi = 0$
 $1 - \cos n\pi = 2$ for odd n
 $\arctan(\frac{1}{\sqrt{3}}) = \frac{\pi}{6}, \arctan(\sqrt{3}) = \frac{\pi}{3}$

$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
 $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$

$\sin x \sin y = \frac{1}{2}[\cos(x - y) - \cos(x + y)]$
 $\cos x \cos y = \frac{1}{2}[\cos(x - y) + \cos(x + y)]$
 $\sin x \cos y = \frac{1}{2}[\sin(x - y) + \sin(x + y)]$

$\sin a \pm \sin b = 2 \sin \frac{a \pm b}{2} \cos \frac{a \mp b}{2}$
 $\cos a + \cos b = 2 \cos \frac{a + b}{2} \cos \frac{a - b}{2}$
 $\cos a - \cos b = -2 \sin \frac{a + b}{2} \sin \frac{a - b}{2}$

$C \cos(\omega_0 t + \theta) = C \cos(\theta) \cos(\omega_0 t) - C \sin(\theta) \sin(\omega_0 t)$
 $C \sin(\omega_0 t + \theta) = C \sin(\theta) \cos(\omega_0 t) + C \cos(\theta) \sin(\omega_0 t)$

$\theta = \tan^{-1}(-\frac{b}{a}), \pm\pi$ when $a < 0$
 $\sin t = \cos(t - \frac{\pi}{2})$
 $-\cos t = \sin(t - \frac{\pi}{2})$

$\cos x = \frac{1}{2}[e^{jx} + e^{-jx}]$
 $\sin x = \frac{1}{2j}[e^{jx} - e^{-jx}]$
 $e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$

$z^* = a - j\mathfrak{b} = re^{-j\theta}$
 $u^*v^* = (uv)^*$

$\angle z = \tan^{-1}(\frac{b}{a}), \pm\pi$ in Q2 and Q3

$z^{\frac{1}{n}} = r^{\frac{1}{n}} e^{j\frac{\theta + 2\pi m}{n}}$

$(s+a)(s+b)(s+c) = s^3 + (a+b+c)s^2 + (ab+bc+ca)s + abc$

Integrals

$\int \cos^2 at \, dt = \frac{t}{2} + \frac{\sin 2at}{4a}$
 $\int t \cos at \, dt = \frac{1}{a^2}(\cos at + at \sin at)$
 $\int t \sin at \, dt = \frac{1}{a^2}(\sin at - at \cos at)$
 $\int t^2 \cos at \, dt = \frac{1}{a^3}(2at \cos at - 2 \sin at + a^2 t^2 \sin at)$
 $\int t^2 \sin at \, dt = \frac{1}{a^3}(2at \sin at + 2 \cos at - a^2 t^2 \cos at)$
 $\int te^{at} \, dt = \frac{1}{a^2}e^{at}(at - 1)$
 $\int t^2 e^{at} \, dt = \frac{1}{a^3}e^{at}(a^2 t^2 - 2at + 2)$
 $\int e^{at} \cos bt \, dt = \frac{1}{a^2 + b^2}e^{at}(a \cos bt + b \sin bt)$
 $\int e^{at} \sin bt \, dt = \frac{1}{a^2 + b^2}e^{at}(a \sin bt - b \cos bt)$
 $\int \frac{1}{x^2 + a^2} \, dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$

Signals

$\mathcal{E}_f = \int_{-\infty}^{\infty} |f(t)|^2 \, dt$ (complex);
 $P_f = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 \, dt$;
rms power = $\sqrt{P_f}$

Cont; analog; periodic (extension); (non/anti)causal;
energy/power (both); deterministic/stochastic (info)

$\int f(t) \cdot \delta(t - t_0) \, dt = f(t_0)$ (f continuous at t_0)

$f(2x - 6)$: shift by 6, scale by 2;
 $f(2(x - 6))$: scale by 2, shift by 6

$f_e(t) = \frac{1}{2}[f(t) + f(-t)]$
 $f_o(t) = \frac{1}{2}[f(t) - f(-t)]$

Systems

$\mathcal{T}: \sum_{k=0} a_k D^k y(t) = \sum_{l=0} b_l D^l f(t)$

Linear $\mathcal{T}[kf_1(t) + f_2(t)] = ky_1(t) + y_2(t)$.
Lin if a_k, b_l are not functions of $y(t), f(t)$
E. $\sin \dot{y}(t) + t^2 y(t) = (t + 3)f(t)$

Time-inv $\mathcal{T}[f(t - \tau)] = y(t - \tau)$.
 a_k, b_l indep of t (const coeff)
Let $g(t) \equiv f(t - \tau)$, find $z(t) = \mathcal{T}[g(t)]$, cmp $y(t - \tau)$

Causal $y(t)$ dep only on $f(\tau), \tau \leq t$. Compare t and τ .

Instantaneous y only dep f at present (no $\int$, no memory)

Invertible given $y(t)$, we can know $f(t)$ (ideal diff is not)

Conv prop

$c(t) \equiv \int_{-\infty}^{\infty} f(\tau)g(t - \tau) \, d\tau$
 $c[n] \equiv \sum_{m=-\infty}^{\infty} f[m]g[n - m]$
 $f * g = g * f$
 $f * (g + h) = f * h + g * h$
 $f * (g * h) = (f * g) * h$
pf: $f * (g * h) = f * (h * g)$
 $= \int f(\tau_1) \int h(\tau_2) g(t - \tau_2 - \tau_1) \, d\tau_2 \, d\tau_1$
 $= \int h(\tau_2) \int f(\tau_1) g(t - \tau_1 - \tau_2) \, d\tau_1 \, d\tau_2$

$f(t - T_1) * g(t - T_2) = c(t - T_1 - T_2)$

$f(at) * g(at) = |\frac{1}{a}| c(at)$ (even/odd)

$f^{(m)}(t) * g^{(n)}(t) = c^{(m+n)}(t)$
pf: $\dot{f}(\tau) = \lim_{T \rightarrow 0} f(\tau) - f(\tau - T)$

Graph: shift **left** by **+t**, and reflect;
Every τ replaced by **t - τ** ; Reverted

Conv table

$f(t) * \delta(t - T) = f(t - T)$

$u(t) * u(t) = t u(t)$

$e^{at} u(t) * u(t) = \frac{1 - e^{at}}{-a} u(t)$

$e^{at} u(t) * e^{bt} u(t) = \frac{e^{at} - e^{bt}}{a - b} u(t)$ $a = b, te^{at} u(t)$

$e^{at} u(t) * e^{bt} u(-t) = \frac{e^{at} u(t) + e^{bt} u(-t)}{b - a}$ $\Re(b) > \Re(a)$

$te^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$

$t^m u(t) * t^n u(t) = \frac{m! n!}{(m+n+1)!} t^{m+n+1} u(t)$

Don't forget $[u(t + T_1) - u(t - T_2)]$ term

LTI response

$Q(D)y(t) = P(D)f(t)$, typically integrating f
Assume causal input $f(t)u(t)$

$y_{zs}(t) = f(t) * h(t)$ from input
 $y_{zs}(0^-) = 0, y_{zs}(0^+) \neq 0$

Let $h(t) = \mathcal{T}[\delta(t)]$ (impulse response)
 $y_{zs}(t) = \mathcal{T}[f(t)] = \mathcal{T}[f(t) * \delta(t)]$
 $= \mathcal{T}[\lim \sum f(n\Delta\tau)\delta(t - n\Delta\tau)\Delta\tau]$
 $= \lim \sum f(\mathfrak{n}\Delta\tau)h(t - \mathfrak{n}\Delta\tau)\Delta\tau = f * h$

$y_{zi}(t)$ from ini, $f(t) = 0, Qy_{zi}(t) = 0$
 $y_{zi}(0^-) = y_{zi}(0^+), \dot{y}_{zi}(0^-) = \dot{y}_{zi}(0^+)$

Ortho set

$\mathcal{E}_e = \int_{t_1}^{t_2} [e(t)]^2 \, dt$
 $= \int_{t_1}^{t_2} f^2(t) \, dt - 2 \sum c_i \int_{t_1}^{t_2} f(t)x_i(t) \, dt + \int_{t_1}^{t_2} (\sum c_i x_i(t))^2 \, dt$
 $= \mathcal{E}_f - 2 \sum_{c_i} \langle f, x_i \rangle$
 $+ (\sum c_i^2 \int_{t_1}^{t_2} x_i(t)^2 \, dt + \sum_{i \neq j} c_i c_j \int_{t_1}^{t_2} x_i(t)x_j(t) \, dt)$

$\frac{\partial \mathcal{E}_e}{\partial c_i} = 0 = -2 \langle f(t), x_i(t) \rangle + 2 \mathcal{E}_i c_i$
 $\mathcal{E}_e^{\min} = \mathcal{E}_f - \sum_{i=1}^N c_i^2 \mathcal{E}_i$

$c_i = \frac{1}{\mathcal{E}_i} \langle f, x_i \rangle = \frac{\int f(t)x_i(t) \, dt}{\int x_i^2(t) \, dt}$

For ortho, $E_z = E_x + E_y$

$|u + v|^2 = |u|^2 + |v|^2 + u^* v + v^* u$

$\langle x(t), y(t) \rangle = \int_{t_1}^{t_2} x(t)y(t)^* \, dt = \int_{t_1}^{t_2} x(t)y(t) \, dt$ if real
Use prod $\rightarrow$ sum identities

FS

$a_0 = \frac{1}{T_0} \int_{T_0} f(t) dt$
 $a_n = \frac{2}{T_0} \int_{T_0} f(t) \cos(n\omega_0 t) dt$
 $b_n = \frac{2}{T_0} \int_{T_0} f(t) \sin(n\omega_0 t) dt$
Energy: T_0 for $n = 0$; $T_0/2$ else

Half wave sym $f(t - \frac{T_0}{2}) = -f(t)$

$a_{n\text{odd}} = \frac{4}{T_0} \int_0^{T_0/2} f(t) \cos(n\omega_0 t) dt$

$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}$
 $F_n = \frac{1}{T_0} \int_{T_0} f(t) e^{-jn\omega_0 t} dt$

$C_n \cos(n\omega_0 t + \theta_n) = \frac{C_n}{2} (e^{j(n\omega_0 t + \theta_n)} + e^{-j(n\omega_0 t + \theta_n)})$
 $= (\frac{C_n}{2} e^{j\theta_n}) e^{jn\omega_0 t} + (\frac{C_n}{2} e^{-j\theta_n}) e^{-jn\omega_0 t}$

$F_n = \frac{C_n}{2} e^{j\theta_n} = \frac{1}{2} (a_n - j b_n) = |F_n| e^{j\angle F_n}$
 $F_{-n} = \frac{C_n}{2} e^{-j\theta_n} = \frac{1}{2} (a_n + j b_n)$

Existence

Weak: finite $\int$, fin bounds a , b , fin power

Strong: fin min/max/discont over T_0 , $\rightarrow \frac{f(t_0^+) + f(t_0^-)}{2}$

FS prop

Time shift $f(t - t_0) \rightarrow F_n e^{-jn(\omega_0 t_0)}$
 $|F_n|$ same; $\angle F_n$ shifted by $-(\omega_0 t_0)n$

Reversal $f(-t) \rightarrow F_{-n}$
Scaling $T = \frac{T_0}{a}$, $\omega = a\omega_0$

Multiplication (same T_0): $f(t)g(t) \rightarrow F_n * G_n$
 $\frac{1}{T_0} \int_{T_0} f(t)g(t) e^{jn\omega_0 t} dt$
 $= \frac{1}{T_0} \int (\sum F_m e^{jm\omega_0 t}) (\sum G_k e^{jk\omega_0 t}) e^{-jn\omega_0 t} dt$
 $= \sum_m \sum_k F_m G_k \frac{1}{T_0} \int_{T_0} e^{j(m+k-n)\omega_0 t} dt$
 $= \sum_m \sum_k F_m G_k \langle e^{j(m+k)\omega_0 t}, e^{jn\omega_0 t} \rangle$
 $= \sum_{k=-\infty}^{\infty} G_k F_{n-k}$

Conjugation $f(t)^* = F_{-n}^*$

Parseval (power sig): $P_f = \frac{1}{T_0} \int_{T_0} f(t)f(t)^* dt$
 $= \frac{1}{T_0} \int_{T_0} (\sum_n F_n e^{jn\omega_0 t}) (\sum_m F_m^* e^{jm\omega_0 t})^* dt$
 $= \sum_n \sum_m F_n F_m^* \frac{1}{T_0} \int_{T_0} e^{j(n-m)\omega_0 t} dt$
 $= \sum_n |F_n|^2 \cdot 1$

f real $\rightarrow |F|$ even, $\angle F$ odd

f real, even $\rightarrow F$ real, even; $F_{-n} = F_n = F_n^*$

f real, odd $\rightarrow F$ imaginary, odd; $-F_{-n} = F_n = -F_n^*$

$f_e(t) \rightarrow \Re\{F_n\}$
 $f_o(t) \rightarrow j \Im\{F_n\}$

Common FS

$(A = 1, T = 2\pi, \omega = 1)$
Square $\frac{4}{\pi} (\cos t - \frac{1}{3} \cos 3t + \frac{1}{5} \cos 5t - \dots)$
 $\frac{4}{\pi} (\sin t + \frac{1}{3} \sin 3t + \frac{1}{5} \sin 5t + \dots)$
Triangle $\frac{8}{\pi^2} (\sin t - \frac{1}{9} \sin 3t + \frac{1}{25} \sin 5t - \dots)$
 $\frac{8}{\pi^2} (\cos t + \frac{1}{9} \cos 3t + \frac{1}{25} \cos 5t + \dots)$
Sawtooth $\frac{2}{\pi} (\sin t - \frac{1}{2} \sin 2t + \frac{1}{3} \sin 3t - \dots)$
 $\frac{2}{\pi} (-\sin t - \frac{1}{2} \sin 2t - \frac{1}{3} \sin 3t - \dots)$
 δ train $\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$
 $\frac{1}{T_0} \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t}$

FT

Let $F(\omega) \equiv \int f(t) e^{-j\omega t} dt$

$F_n = \frac{1}{T_0} \int_{T_0} f(t) e^{-jn\omega_0 t} dt$

Limit as $\omega_0 = \Delta\omega \rightarrow 0$,
 $F_n = \frac{\Delta\omega}{2\pi} \int_{-\infty}^{\infty} f(t) e^{-jn\Delta\omega t} dt \equiv \frac{\Delta\omega}{2\pi} F(n\Delta\omega)$

$f_{T_0}(t) = \sum F_n e^{jn\omega_0 t} = \sum \frac{\Delta\omega}{2\pi} F(n\Delta\omega) e^{jn\Delta\omega t}$

$f(t) = \lim_{T_0 \rightarrow \infty} f_{T_0}(t) = \frac{1}{2\pi} \int F(\omega) e^{j\omega t} d\omega$

$F(\omega) = |F(\omega)| e^{j\angle F(\omega)}$

Real signals: amp and phase symmetry

Existence: weak: energy signal ($|e^{-j\omega t}| = 1$)

Strong: fin num max/min/discont

FT Table

$\delta(t) \rightarrow 1$
 $1 \rightarrow 2\pi\delta(\omega)$
 $e^{j\omega_0 t} \rightarrow 2\pi\delta(\omega - \omega_0)$
 $\cos\omega_0 t \rightarrow \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
 $\sin\omega_0 t \rightarrow j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$

$\sum \delta(t - nT_0) \rightarrow \omega_0 \sum \delta(\omega - n\omega_0)$

$e^{-at} u(t) \rightarrow \frac{1}{a+j\omega}$
 $e^{-a|t|} \rightarrow \frac{2a}{a^2+\omega^2}$
 $u(t) = \lim_{a \rightarrow 0} e^{-at} u(t) \rightarrow \lim \frac{1}{a+j\omega}$
 $= \lim (\frac{a}{a^2+\omega^2} - j \frac{\omega}{a^2+\omega^2}) = \pi\delta(\omega) + \frac{1}{j\omega}$
 $\text{sgn}(t) \rightarrow \frac{2}{j\omega}$

$t^n e^{-at} u(t) \rightarrow \frac{n!}{(a+j\omega)^{n+1}}$

$\cos\omega_0 t u(t) \rightarrow \frac{\pi}{2} (\delta(\omega - \omega_0) + \delta(\omega + \omega_0)) + \frac{j\omega}{\omega_0^2 - \omega^2}$
 $\sin\omega_0 t u(t) \rightarrow \frac{\pi}{2j} (\delta(\omega - \omega_0) - \delta(\omega + \omega_0)) + \frac{\omega_0}{\omega_0^2 - \omega^2}$
 $e^{-at} \cos\omega_0 t u(t) \rightarrow \frac{a+j\omega}{(a+j\omega)^2 + \omega_0^2} \quad a > 0$
 $e^{-at} \sin\omega_0 t u(t) \rightarrow \frac{\omega_0}{(a+j\omega)^2 + \omega_0^2} \quad a > 0$

$\text{rect}(\frac{t}{\tau}) \rightarrow \tau \text{sinc}(\frac{\tau}{2}\omega)$
 $\frac{W}{\pi} \text{sinc}(Wt) \rightarrow \text{rect}(\frac{\omega}{2W})$

$\Delta(\frac{t}{\tau}) \rightarrow \frac{\tau}{2} \text{sinc}^2(\frac{\tau}{4}\omega)$
 $\frac{W}{2\pi} \text{sinc}^2(\frac{W}{2}t) \rightarrow \Delta(\frac{\omega}{2W})$

$[\omega^2 \text{r}(\frac{\omega}{2\omega_0})] \leftarrow \frac{1}{2\pi} \frac{e^{j\omega t}}{(jt)^3} (-\omega^2 t^2 - 2j\omega t + 2)_{-\omega_0}^{\omega_0}$
 $= \frac{(\omega_0^2 t^2 - 2) \sin\omega_0 t + 2\omega_0 t \cos\omega_0 t}{\pi t^3}$

$[\frac{|\omega|}{\omega_0} \text{rect}(\frac{\omega}{2\omega_0})] \leftarrow \frac{\cos\omega_0 t + \omega_0 t \sin\omega_0 t - 1}{\omega_0 \pi t^2}$

Frequency domain prop

Linearity

Time shift $f(t - t_0) \rightarrow F(\omega) e^{-jt_0\omega}$
 $|F|$ unchanged; $\angle F = -t_0\omega$, lin shift

Freq shift $f(t) e^{j\omega_0 t} \rightarrow F(\omega - \omega_0)$

Duality $f(t) \rightarrow F(\omega)$, $F(t) \rightarrow 2\pi f(-\omega)$
pf. $f(t) = \frac{1}{2\pi} \int F(\lambda) e^{j\lambda t} d\lambda$
 $2\pi f(-t) = \int F(\lambda) e^{-tj\lambda} d\lambda = \mathcal{F}[F(\lambda)]$

Reversal $f(-t) \rightarrow F(-\omega)$

Scaling $f(at) \rightarrow \frac{1}{|a|} F(\frac{\omega}{a})$

Convolution $f * g \rightarrow FG$, $fg \rightarrow \frac{1}{2\pi} F * G$
 $\mathcal{F}[f * g] = \int e^{-j\omega t} \int f(\tau) g(t - \tau) d\tau dt$
 $= \int f(\tau) \mathcal{F}[g(t - \tau)] d\tau$
 $= \int f(\tau) G(\omega) e^{-j\omega\tau} d\tau$
 $\frac{1}{2\pi} \mathcal{F}^{-1}[F * G] = (\frac{1}{2\pi})^2 \int e^{j\omega t} \int F(\lambda) G(\omega - \lambda) d\lambda d\omega$

Diff $f^{(n)}(t) \rightarrow (j\omega)^n F(\omega)$ (diff $e^{j\omega t}$)

Int $\int_{-\infty}^t f(\tau) d\tau \rightarrow \frac{1}{j\omega} F(\omega) + \pi F(0)\delta(\omega)$
 $\int = f(t) * u(t) \rightarrow F(\omega) U(\omega)$
 $U(\omega) = \lim \frac{1}{a+j\omega} = \lim (\frac{a}{a^2+\omega^2} - j \frac{\omega}{a^2+\omega^2})$
 $= \pi\delta(\omega) + \frac{1}{j\omega} \quad (\int \frac{a}{\omega^2+a^2} d\omega = \tan^{-1} = \pi)$

Conjugation $f(t)^* \rightarrow F(-\omega)^*$

Symmetry $\text{Re} \rightarrow \text{mag even}$, phase odd ($F(-\omega) = F(\omega)^*$)
real, even $\rightarrow$ real, even; real, odd $\rightarrow$ imaginary, odd

f even: $F(\omega) = 2 \int_0^{\infty} f(t) \cos(\omega t) dt$
 f odd: $F(\omega) = -2j \int_0^{\infty} f(t) \sin(\omega t) dt$

$\omega_0 = \frac{2\pi}{T_0}$

$a > 0$

$a > 0$

$a > 0$

Parseval

$E_f = \int |f(t)|^2 dt = \frac{1}{2\pi} \int |F(\omega)|^2 d\omega$ for energy sig
 $= \int f f^* dt = \int f(t) \mathcal{F}^{-1}[F(-\omega)^*] dt$
 $= \int f(t) \frac{1}{2\pi} \int F(-\omega)^* e^{j\omega t} d\omega dt$
 $= \frac{1}{2\pi} \int f(t) \int F(\lambda)^* e^{-j\lambda t} d\lambda dt$
 $= \frac{1}{2\pi} \int F(\lambda)^* \int f(t) e^{-j\lambda t} dt d\lambda$

$\Delta E_f = \frac{2}{2\pi} \int_{\omega_1}^{\omega_2} |F(\omega)|^2 d\omega$
Autocorrelation $\psi_f(t) \equiv \int f(\tau) f(\tau - t) d\tau \rightarrow |F(\omega)|^2$

Modulation

$m(t) \cos(\omega_c t) \rightarrow \frac{1}{2} [M(\omega + \omega_c) + M(\omega - \omega_c)]$
 $e(t) = m(t) \cos^2 \omega_c t$
 $E(\omega) = \frac{1}{2} M + \frac{1}{4} [M(\omega + 2\omega_c) + M(\omega - 2\omega_c)] \rightarrow \text{LPF}$

SSB 1/4 gain

$\phi_{AM}(t) = [A + f(t)] \cos(\omega_0 t)$
 $A \geq f(t)$ for all t
modulation index $\mu \equiv f_{\max}/A$
 $\mu = \infty$, suppressed carrier; $\mu = 1$, marginal

LTIC sys trans, (marginally) stable

Let $e^{j\omega t} \Rightarrow H(\omega) e^{j\omega t}$
 $\lim \sum \frac{F(n\Delta\omega)\Delta\omega}{2\pi} e^{jn\Delta\omega t} \Rightarrow \lim \sum \frac{F(n\Delta\omega)H(n\Delta\omega)\Delta\omega}{2\pi} e^{jn\Delta\omega t}$
 $y(t) = \frac{1}{2\pi} \int F(\omega) H(\omega) e^{j\omega t} d\omega$
 $Y(\omega) = F(\omega) H(\omega)$

Distortionless $y(t) = k f(t - t_d)$, so $H(\omega) = k e^{-j\omega t_d}$

Payley-Wiener H realizable, h causal iff

$\int \frac{|\ln |H(\omega)||}{1+\omega^2} d\omega < \infty$ (consecutive 0s)
Truncate $\hat{h}(t) = h(t) u(t)$

Periodic FT

$f(t) = \sum_n F_n e^{jn\omega_0 t}$
 $\mathcal{F}[f(t)] = 2\pi \sum_n F_n \delta(\omega - n\omega_0)$

$Y = F(\omega) H(\omega) = 2\pi \sum F_n H(n\omega_0) \delta(\omega - n\omega_0)$
 $Y_n \equiv F_n H(n\omega_0)$. Periodic with same ω_0

Eigen: $f(t) = e^{j\omega_0 t}$, $Y_1 = H(1\omega_0)$, $y(t) = H(1\omega_0) e^{j\omega_0 t}$

$f(t) = \cos(\omega_0 t + \theta)$, assume $h(t)$ real
 $y = \frac{1}{2} (e^{j(\theta+\omega_0 t)} H(\omega_0) + e^{-j(\theta+\omega_0 t)} H(-\omega_0))$
 $= |H(\omega_0)| \cos(\omega_0 t + \theta + \angle H(\omega_0))$

$\cos 2t * e^{-3t} u(t) \equiv f * h$
 $= |H(2)| \cos(2t + \angle H(2))$

Sampling

$\bar{f}(t) \equiv f(t) \delta_{T_s}(t) = \sum f(nT_s) \delta(t - nT_s)$
 $\bar{F}(\omega) = \frac{1}{2\pi} F(\omega) * [\frac{2\pi}{T_s} \sum \delta(\omega - n\omega_s)] = \frac{1}{T_s} \sum F(\omega - n\omega_s)$
 $\omega_s \geq 4\pi B$, $F_s \geq F_N \equiv 2B$

Intrapolation when $F_s = 2B$

$F(\omega) = \bar{F}(\omega) T_s \text{rect}(\frac{\omega}{4\pi B})$
If $F_s = 2B$, $f(t) = \bar{f}(t) * \frac{2B}{F_s} \text{sinc}(2\pi B t)$
 $= \sum_n f(nT_s) \delta(t - nT_s) * \text{sinc}(2\pi B t)$
 $= \sum_n f(nT_s) \text{sinc}(2\pi B t - n\pi)$
ana FS, with basis sinc, weighted sample sum

If $F_s > 2B$, $f(t) = \sum f(nT_s) w(t - nT_s)$ not sinc weight
for some relaxed LP filter $w(t)$

Anti-alias before sampling: LPF of $F_s/2$

Practical sampling

$p_T(t) = \frac{\tau}{T_s} + \sum (\frac{2}{\pi n} \sin(n\pi \frac{\tau}{T_s})) \cos(n\omega_s t)$
 $P_T(\omega) = 2\pi \frac{\tau}{T_s} \delta(\omega) + \sum \frac{\pi 2 \sin(...)}{\pi n} [\delta(\omega + n\omega_s) + \delta(\omega - n\omega_s)]$

LT

$\mathcal{L}[-e^{-at} u(-t)] = \mathcal{L}[e^{-at} u(t)]$, except ROC
If sig are causal, $\mathcal{L}$ is 1-to-1

Unilateral: $\mathcal{L}[f] = \int_0^\infty f(t) e^{-st} dt$
 $= \int [f(t) e^{-\sigma t}] e^{-j\omega t} dt$

σ_0 : smallest σ to make integral converge

Uni LT Table

Watch ROC!!
 $\delta(t) \rightarrow 1$
 $u(t) \rightarrow \frac{1}{s}$
 $t^n u(t) \rightarrow \frac{n!}{s^{n+1}}$

$e^{\lambda t} u(t) \rightarrow \frac{1}{s-\lambda}$ $\Re(s) > \Re(\lambda)$
 $t^n e^{\lambda t} u(t) \rightarrow \frac{n!}{(s-\lambda)^{n+1}}$
 $\frac{1}{(n-1)!} t^{n-1} e^{\lambda t} u(t) \rightarrow \frac{1}{(s-\lambda)^n}$
 $e^{-at} \cos(bt) u(t) \rightarrow \frac{s+a}{(s+a)^2 + b^2}$
 $e^{-at} \sin(bt) u(t) \rightarrow \frac{b}{(s+a)^2 + b^2}$

$re^{-at} \cos(bt + \theta) u(t) \rightarrow \frac{(r \cos \theta) s + (ar \cos \theta - br \sin \theta)}{s^2 + 2as + (a^2 + b^2)}$
 $2re^{-at} \cos(bt + \theta) u(t) \rightarrow \frac{re^{j\theta}}{s - (-a + jb)} + \frac{re^{-j\theta}}{s - (-a - jb)}$
 $e^{-at} [A \cos(bt) + \frac{B - Aa}{b} \sin(bt)] u(t)$
 $= \frac{\sqrt{A^2 c + B^2 - 2ABa}}{b} e^{-at} \cos\left(bt + \tan^{-1} \frac{Aa - B}{Ab}\right) u(t)$
 $\rightarrow \frac{As + B}{s^2 + 2as + c} \quad b \equiv \sqrt{c - a^2}$

LT Prop

Linearity ROC $\cap$
Time delay $f(t - t_0) u(t - t_0) \rightarrow F(s) e^{-st_0}$ ROC same
 $t_0 > 0$; pf: $\int_{-t_0}^\infty$
Freq shift $f(t) e^{s_0 t} \rightarrow F(s - s_0)$ $\Re(s) > \sigma_0 + \Re(s_0)$
Scaling ($a > 0$), $f(at) \rightarrow \frac{1}{|a|} F(\frac{s}{a})$ $\Re(s) > a\sigma_0$
Convolution $f_1 * f_2 \rightarrow F_1 F_2$ ROC $\cap$
 $f_1 f_2 \rightarrow \frac{1}{2\pi j} F_1 * F_2$

Time diff $\dot{f}(t) \rightarrow sF(s) - f(0^-)$ $\Re(s) > \max(\sigma_0, 0)$
 $\ddot{f}(t) \rightarrow s^2 F(s) - s f(0^-) - \dot{f}(0^-)$
pf (parts): $\int_0^\infty \dot{f}(t) e^{-st} dt = [f(t) e^{-st}]_0^\infty + sF(s)$
Time int $\int_0^\infty f(\tau) d\tau \rightarrow \frac{1}{s} F(s)$ pf: diff

Freq diff $-t f(t) \rightarrow \frac{dF(s)}{ds}$
Freq int $\frac{1}{t} f(t) \rightarrow \int_s^\infty F(z) dz$

IVT $f(0^+) = \lim_{s \rightarrow \infty} sF(s)$ if exists
pf: $\mathcal{L}[f(t)] = \int_0^\infty \dot{f}(t) e^{-st} dt$
 $sF(s) - f(0^-) = \int_0^{0^+} \dot{f}(t) e^{-st} dt + \int_{0^+}^\infty \dot{f}(t) e^{-st} dt$
 $sF(s) - f(0^-) = f(0^+) - f(0^-)$
FVT $f(\infty) = \lim_{s \rightarrow 0} sF(s)$ if exists
pf: $\mathcal{L}[\dot{f}(t)] = \int_0^\infty \dot{f}(t) e^{-st} dt$
 $\lim_{s \rightarrow 0} sF(s) - f(0^-) = \lim_{s \rightarrow 0} \int_{0^-}^\infty \dot{f}(t) e^{-st} dt$
 $sF(s) - f(0^-) = f(\infty) - f(0^-)$

Rational $\mathcal{L}^{-1}$

First rationalize

$F(s) = \frac{b_m s^m + \dots + b_1 s + b_0}{1 s^n + \dots + a_1 s + a_0} \equiv \frac{P(s)}{Q(s)}$
 $= \frac{a_0}{(s-\lambda)^r} + \dots + \frac{a_{r-1}}{s-\lambda} + \frac{k_1}{s-\lambda_1} + \dots$
 $k = (s - \lambda_i) F(s)|_{s=\lambda_i}$

$a_0 = (s - \lambda)^r F(s)|_{s=\lambda}$
 $a_m = \frac{1}{m!} \frac{d^m}{ds^m} [(s - \lambda)^r F(s)]|_{s=\lambda}$

$\mathcal{L}^{-1}[\frac{1}{(s-\lambda)^n}] = \frac{1}{(n-1)!} t^{n-1} e^{\lambda t} u(t)$

Multiply both by s and let $s = \infty$, only $\frac{1}{s}$ term left

Sys Anal

$Q(D)y(t) = P(D)f(t)$
 $H(s) = \frac{Y_{zs}(s)}{F(s)} = \frac{P(s)}{Q(s)}$

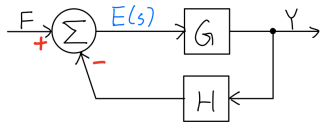
Asym. (internal, init): $y_{zi}(t) \rightarrow 0$ as $t \rightarrow \infty$
marginal: $y_{zi}(t)$ remains bounded (unique λ s on Im axis)

BIBO (external, input) iff $\int_{-\infty}^{\infty} |h(t)| dt$ exists
 $\Rightarrow |f(t)| < K$
 $y_{zs}(t) = h * f = \int h(\tau)f(t-\tau) d\tau$
 $\leq \int |h(\tau)||f(t-\tau)|d\tau < K \int |h(\tau)| d\tau$
 $\Leftarrow$ Let $f(t) = \text{sgn}(h(-t))$
 $y(0) = \int h(\tau)f(0-\tau)d\tau$
 $= \int h(\tau)\text{sgn}(h(\tau))d\tau$
 $= \int |h(\tau)|d\tau \equiv \infty$

Asy stable $\Rightarrow$ BIBO stable (all exp)
Marginal $\Rightarrow$ BIBO unstable ($\int |\sin(t)| dt = \infty$)

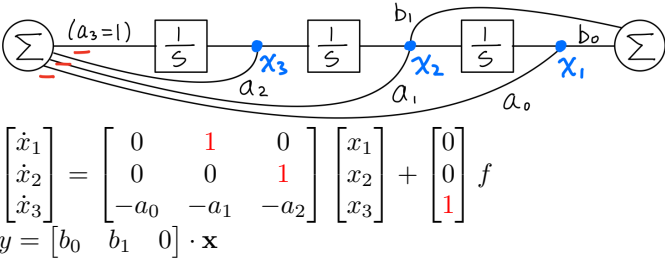
Sys Realization

Feedback: $Y = GE = G(F - HGY)$
 $H_{\text{eff}} = \frac{Y}{F} = \frac{G}{1+HG}$



Canonical

$F(s) = 1s^3X + a_2s^2X + a_1sX + a_0X$
 $s^3X = F - a_2s^2X - a_1sX - a_0X$
 $b_3s^3X + b_2s^2X + b_1sX + b_0X = Y(s)$
 $H = \frac{b_1s+b_0}{1s^3+a_2s^2+a_1s+a_0}$



$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} f$
 $y = \begin{bmatrix} b_0 & b_1 & 0 \end{bmatrix} \cdot \mathbf{x}$

Second form:
 $s^2Y = s^2b_2F + s(-a_1Y + b_1F) + (-a_0Y + b_0F)$
 $Y = b_2F + \frac{1}{s}(-a_1Y + b_1F) + \frac{1}{s^2}(-a_0Y + b_0F)$
Transpose **A**, swap and trans **b**, **c**

Cascade (made $P_3(s)$ constant here)

$H = \frac{P_1(s)}{s+\lambda_1} \cdot \frac{P_2(s)}{s+\lambda_2} \cdot \frac{k_3}{s+\lambda_3}$

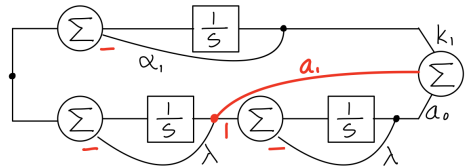
$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -\lambda_1 & 0 & 0 \\ .. & -\lambda_2 & 0 \\ .. & .. & -\lambda_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ .. \\ .. \end{bmatrix} f$
 $y = \begin{bmatrix} 0 & 0 & k_3 \end{bmatrix} \cdot \mathbf{x}$

Parallel

$H = \frac{k_1}{s+\lambda_1} + \frac{k_2}{s+\lambda_2} + \frac{k_3}{s+\lambda_3}$

$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -\lambda_1 & 0 & 0 \\ 0 & -\lambda_2 & 0 \\ 0 & 0 & -\lambda_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} f$
 $y = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} \cdot \mathbf{x}$

(same pole) $H = \frac{a_0}{(s-\lambda)^2} + \frac{a_1}{s-\lambda} + \frac{k_1}{s-\alpha_1}$



State Equations

Def: *state* of a sys at any time t_0 is the *smallest* set of nums $\{x_i(t_0)\}$ that is sufficient to determine sys behavior $\forall t > t_0$ when input $f(t)$ is known for $t > t_0$
Always $\int$ output
If $t_0 = 0$, initial condition

$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}f$
 $y = \mathbf{c} \cdot \mathbf{x}$
MIMO
 $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}f$
 $y = \mathbf{C}\mathbf{x} + \mathbf{D}f$

$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}F(s)$
 $(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{x}(0) + \mathbf{B}F(s)$
 $\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}F(s)$
 $\mathbf{x}(t) = \mathbf{x}_{zi}(t) + \mathbf{x}_{zs}(t)$

LTI Freq Response

$H(\omega) = H(s)|_{s=j\omega} = \frac{P(s)}{Q(s)}$
Ideal delay: $|H| = 1, \angle H(\omega) = -\omega T$
Ideal diff: $|H(\omega)| = |\omega|, \angle H(\omega) = \pm \frac{\pi}{2}$
Ideal int: $|H(\omega)| = \frac{1}{|\omega|}, \angle H(\omega) = \mp \frac{\pi}{2}$

$f(t) = C = Ce^{0t}$
 $y(t) = H(0)C$

$f(t) = e^{st}$ everlasting
 $y(t) = h(t) * e^{st} = e^{st} \int h(\tau)e^{-s\tau}d\tau \equiv H(s)e^{st}$

$f(t) = e^{j\omega_0 t} u(t)$
 $Y_{zs}(s) = F(s)H(s)$
 $= \frac{1}{s-j\omega_0} \frac{P(s)}{(s-\lambda_1) \dots (s-\lambda_n)}$
 $= \frac{H(s)|_{s=j\omega_0}}{s-j\omega_0} + \frac{k_1}{s-\lambda_1} + \dots$
 $y_{zs}(t) = H(\omega_0)e^{j\omega_0 t} u(t) + \sum_i k_i e^{\lambda_i t} u(t)$

$y_{ss}(t)$ scaled input
 $y_{tr}(t)$ decays for stable sys

$f(t) = \cos(\omega_0 t + \theta) u(t)$
 $y_{ss}(t) = |H(\omega_0)| \cos(\omega_0 t + \theta + \angle H(\omega_0)) u(t)$

Pole Zero

$H(j\omega) = b_n \frac{(j\omega-z_1) \dots (j\omega-z_n)}{(j\omega-p_1) \dots (j\omega-p_n)}$
 $\equiv b_n \frac{(r_1 e^{j\phi_1}) \dots (r_n e^{j\phi_n})}{(d_1 e^{j\theta_1}) \dots (d_n e^{j\theta_n})}$
 $|H(\omega)| = b_n \frac{r_1 \dots r_n}{d_1 \dots d_n}$
 $\angle H(\omega) = (\phi_1 + \dots + \phi_n) - (\theta_1 + \dots + \theta_n)$

Vector from p/z to $j\omega$
Pole enhances gain; Zero suppresses